

Lecture - 14

$$\theta \xrightarrow{p(x|\theta)} x \xrightarrow{\text{learning models}} \hat{\theta}$$

$$\textcircled{i} \text{ max } p(x|\theta) \rightarrow \hat{\theta}_{ML}$$

$$\textcircled{ii} p(\theta|x) \rightarrow \hat{\theta}_{MAP}$$

ML estimation for probabilistic linear regression

* n training examples $\{\vec{x}_i, y_i\}_{i=1}^n$
 $\vec{x}_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}$

$$1 \quad y_i = \vec{\theta}^T \vec{x}_i + \epsilon_i \quad \left| \begin{array}{l} \epsilon_i = \text{Gaussian noise} \\ \epsilon_i \sim \mathcal{N}(0, \sigma^2) \end{array} \right.$$

$$\boxed{y_i | \vec{x}_i, \theta \sim \mathcal{N}(\theta^T \vec{x}_i, \sigma^2)}$$

$$\begin{aligned} E[y_i] &= E[\theta^T \vec{x}_i + \epsilon_i] \\ &= E[\theta^T \vec{x}_i] + E[\epsilon_i] \\ &= \theta^T \vec{x}_i + 0 \\ &\rightarrow \text{Var}(y_i) = \text{Var}(\theta^T \vec{x}_i + \epsilon_i) \\ &= \text{Var}(\theta^T \vec{x}_i) + \text{Var}(\epsilon_i) \\ &= \sigma^2 \end{aligned}$$

univariate Gaussian Distribution

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x-\mu)^2}{2\sigma^2} \right\}$$

Multivariate Gaussian Distribution:-

Distribution over a multivariate Random Variable $\vec{x} \in \mathbb{R}^d$ of real numbers defined by a mean vector $\vec{\mu} \in \mathbb{R}^d$ and a covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$

$$P(\vec{x}; \vec{\mu}, \Sigma) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp \left\{ -\frac{1}{2} (\vec{x} - \vec{\mu})^T \Sigma^{-1} (\vec{x} - \vec{\mu}) \right\}$$

$$P(y_i | \vec{x}_i, \vec{\theta}) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp \left\{ -\frac{(y_i - \vec{\theta}^T \vec{x}_i)^2}{2\sigma^2} \right\}$$

Assuming iid responses :-

$$P(\vec{y} | \vec{x}, \vec{\theta}) = \prod_{i=1}^n P(y_i | \vec{x}_i, \vec{\theta})$$

$$= \left(\frac{1}{\sqrt{2\pi}\sigma^2} \right)^{n/2} \exp \left\{ -\sum_{i=1}^n \frac{(y_i - \vec{\theta}^T \vec{x}_i)^2}{2\sigma^2} \right\}$$

$$\log P(\vec{y} | \vec{x}, \vec{\theta}) \propto - \underbrace{\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \vec{\theta}^T \vec{x}_i)^2}_{\text{proportional to}}$$

$$- \log P(\vec{y} | \vec{x}, \vec{\theta}) \propto \underbrace{\frac{1}{2} \sum_{i=1}^n (y_i - \vec{\theta}^T \vec{x}_i)^2}_{\text{Least Square Error.}}$$

MLE will give the same solution as unregularized

least squares.

MAP Estimation for Probabilistic Linear Regression

$$p(\vec{\theta}) = \mathcal{N}(\underbrace{0}_{\text{mean}}, \underbrace{\sigma_{\theta}^2 I}_{\text{cov}}) \propto \exp \left\{ -\frac{\vec{\theta}^T \vec{\theta}}{2\sigma_{\theta}^2} \right\}$$

log likelihood

$$\log p(\vec{y} | \vec{x}, \vec{\theta}) \propto -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \vec{\theta}^T \vec{x}_i)^2$$

The MAP objective will be the log-likelihood
 $\uparrow$
 $\log p(\theta)$

$$J(\theta) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \vec{\theta}^T \vec{x}_i)^2 - \frac{\vec{\theta}^T \vec{\theta}}{2\sigma_{\theta}^2}$$

maximizing $J(\theta)$ is equivalent to minimizing

$$\mathcal{L}(\theta) = \sum_{i=1}^n (y_i - \vec{\theta}^T \vec{x}_i)^2 + \frac{\sigma^2}{\sigma_{\theta}^2} \vec{\theta}^T \vec{\theta}$$

$$\lambda = \frac{\sigma^2}{\sigma_{\theta}^2} \rightarrow \text{Equivalent to ridge regression.}$$

- Summary:-
- (i) MLE $\rightarrow$ unregularized soln
 - (ii) MAP $\rightarrow$ regularized soln
 - (iii) Gaussian likelihood $\rightarrow$ Squared loss
 - (iv) Gaussian prior $\rightarrow$ L_2 regularization.